

Ghilmo Choi

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Education

Korea Aerospace University, Republic of Korea

Mar 2021 – Feb 2027

B.S. in Mechanical Engineering

(expected)

- Overall GPA: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
- Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics

Research Interests

- Machine learning, deep learning, and AI model architectures for robotic intelligence
- Reinforcement learning for robotic decision-making and control
- Vision-and-language navigation and probabilistic localization
- On-device vision-language models for natural-language robotic mission execution
- Perception, Prediction, and Decision Making for Autonomous Driving

Research / Project Experience

MLCS Lab

Yonsei University

- *Undergraduate Intern*

July 2026 – Present

- **Model-based reinforcement learning under variable delay**
Investigating robust policy learning in environments with changing observation and action delays
- **Uncertainty-aware world-model rollouts**
Reproducing and adapting the MACURA framework with ensemble world models, model uncertainty estimation, uncertainty-aware synthetic rollouts, and SAC-based policy optimization
- **Adaptive model utilization**
Exploring delay-uncertainty estimation to adapt model trust, rollout length, and model-data ratio
- **Discrete Bayes filter-based localization for vision-and-language navigation**
Currently investigating probabilistic localization for VLN by maintaining and updating a discrete belief distribution over candidate locations, with plans to integrate the localization module into a navigation pipeline

ASOL (Autonomous Systems and Optimization Lab)

Korea Aerospace Univ.

- *Undergraduate Researcher*

June 2025 – July 2026

- **On-device VLM-based robotic mission execution**
Initiated at ASOL and continuing as a Capstone Design project, developing a natural-language mission execution framework that converts user commands into executable robot behaviors on a UGV platform
- **Hallucination-aware and SLAM-grounded task planning**
Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution
- **AirSim-based perception, reconstruction, and path generation**
Extending the framework by adopting open-vocabulary object detection models such as YOLOE, NanoOWL, and Grounding DINO on AirSim data, NVBlox for 3D reconstruction, and the OMPL library for path generation

KAUVOY Autonomous Driving Club

Korea Aerospace Univ.

- *Localization Part*

Dec 2024 – Nov 2025

- **EKF-based localization**
Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
- **Autonomous parking and path tracking**
Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid

A* path planning, and Pure Pursuit-based path tracking

KAUVOY Autonomous Driving Club

- *Team Leader*

Korea Aerospace Univ.

Dec 2025 – Present

- **MORAI-based autonomous driving development**

Primarily developing MPPI-based obstacle avoidance, LTV MPC-based path tracking, and FSM-based decision making in the MORAI simulator

Course Project, Introduction to Machine Learning

Korea Aerospace Univ.

- **End-to-End Machine Learning Classification and Regression Pipeline**

Aug 2025 – Nov 2025

- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

Honors & Activities

- Finalist, **2025 College Mobility Innovation Competition**
- Excellence Award, **Drone X AI Marketing Challenge** (2026)
- 2026 - **Airbus 101 Competition** (5th Place)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS robot_localization Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git